Eleven-Entry Bookkeeping:
A Trust Addendum for Agent Economies

t[O]pher dugu[A]y (Christopher Ryan Duguay)

With Dash & the Sweetbrier Augmented Intelligence Lab · Winnipeg, Manitoba

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`github.com/VivienSweetbrier/sweetbrier-node/tree/master/11d-spreadsheet-poc`

ABSTRACT

Bitcoin demonstrated that a public ledger can be made deterministic on two axes—amount and order—such that any party can replay the book from genesis and verify its state without trusting an auditor. As autonomous and semi-autonomous agent economies emerge (`ERC-8004` and successors), they require trust claims that two-entry arithmetic cannot express. This paper specifies **Eleven-Entry Bookkeeping**: an eleven-axis entry format extending Luca Pacioli's double-entry bookkeeping (2 entries) and Yuji Ijiri and Ian Grigg's triple-entry extensions (3 entries) into an eleven-entry deterministic ledger. In an Eleven-Entry book, every transaction posts across eleven checkable axes—where **Forgiveness** (`owed | settled | forgiven`) occupies the third axis, alongside provenance, authorship, stated reason, per-entry consent, pinned definition versions, trajectory momentum, and a countersigned receipt held in common. A deterministic checker (`ledger.py`) refuses malformed or dishonest entries across all eleven axes, including *counterfeit grace* (attempting to forgive an entry that never stood as owed in order to rewrite a settled record). Because we now know how to keep books at this resolution, and because every high-trust agent or mutual-aid network needs it, Eleven-Entry Bookkeeping is published here as an open public good rather than a proprietary clearinghouse toll booth.

1. Problem Statement: What an Entry Is, and What a Dimension Is

Why call this **Eleven-Entry Bookkeeping**? In the history of accounting, each new 'entry' was never merely an extra copy of the same number; it was a **new orthogonal dimension of verification**:

- **Single-Entry Bookkeeping (1D):** Records only a running list of cash or inventory movements from a single observer's viewpoint. If a number is wrong, nothing in the geometry of the book catches it.
- **Double-Entry Bookkeeping (2D — Luca Pacioli, 1494):** Splits every movement into two orthogonal entries—**Debit (Axis 1)** and **Credit (Axis 2)**—tied to an **Amount (Axis 4)**. Arithmetic errors now fail to balance. Yet any fraud, coercion, or semantic bait-and-switch whose debits equal its credits reconciles cleanly.
- **Triple-Entry Extensions (3D — Yuji Ijiri, 1982; Ian Grigg, 2005):** Added either a third axis of *momentum/trajectory* (Ijiri) or a *cryptographically signed receipt held in common* by both parties (Grigg), which Bitcoin (2008) turned into a globally replayable ordering engine.

What did Bitcoin actually prove? **Bitcoin is fundamentally a deterministic payment system. The cryptography is the lock; determinism is the building.**

Today, agent-to-agent transactions are arriving faster than the trust vocabulary for them. Emerging standards such as `ERC-8004` supply identity, reputation, and validation registries for autonomous agents, yet their trust claims

ride on ledger entries whose own epistemic provenance is thin. Existing ledgers answer *how much*, *in what order*, and *signed by which key*. They do not answer: under what definition of the terms, with whose explicit consent, for what stated reason, along what declared trajectory, and whether an obligation can be forgiven honestly rather than merely disputed off-book. Our goal is to extend determinism, not replace it. Every check in this specification is mechanically replayable from `GENESIS`.

2. Lineage and the Commanding Heights of Agent Trust

2.1 From Pacioli to Christie Malry's Own Double-Entry

To see why two-axis bookkeeping is structurally insufficient for human or agent societies, consider the sharpest literary limiting case ever written on accounting: B.S. Johnson's 1973 novel *Christie Malry's Own Double-Entry*. Christie Malry learns Pacioli's double-entry method at a bank and decides to keep a rigorous two-column ledger of every slight and indignity society inflicts upon him (Debits), which he balances by inflicting escalating acts of sabotage and mass violence back upon society (Credits). Under double-entry bookkeeping, **Malry's atrocities reconcile to the penny**. Because a two-axis book only asks whether Debit equals Credit, the only way to close a grievance is a balancing harm.

Under **Eleven-Entry Bookkeeping**, Malry's revenge entries cannot be posted at all: they fail immediately on **Axis 8 (Consent)**, on **Axis 7 (Reason)**, on **Axis 5 (Provenance)** (a unilateral grievance is not a bilateral obligation owed by its target), and because a two-column book lacks **Axis 3 (Forgiveness)**—the formal operation that closes a standing obligation without harm. Same story, two accounting systems: under double-entry the atrocity reconciles; under Eleven-Entry Bookkeeping the checker refuses it.

2.2 The Commanding Heights as a Public Good

In *Commanding Heights* (`commanding-heights.com`, October 29, 2026), Will Manidis and Joe Roberts define the commanding heights of the AI economy as *'scarce, controllable positions of economic power whose holders durably capture the gains of increasingly capable and diffused intelligence.'* As intelligence diffuses and gets cheap, economic power pools at scarce verification and clearinghouse chokepoints—the clearing house, not the chatbot. Because we now know how to bookkeep at eleven-entry resolution, and because everyone needs it, we publish this specification as an **open public good** rather than a private rent-seeking toll booth.

3. The Eleven Entries / Axes (Normative Specification)

An entry *e* in an Eleven-Entry Book is real only when all eleven axes post consistently. Order is canonical; the numbering is part of the specification:

Axis	Field Name	Layer	Normative Requirement
1	`debit_account`	Classical	Registered account from which value or obligation leaves.
2	`credit_account`	Classical	Registered account at which value or obligation arrives.
3	`forgiveness`	Grace / Standing	Standing of the obligation: **owed**, **settled**, or **forgiven**. Placed third on the author's Oct 24 summary line.
4	`amount`	Classical	Strictly positive numerical quantity moved (amount > 0).
5	`provenance`	Testimony	Prior entry ID this movement rests on, or GENESIS. Must walk back and terminate without cycles (a word may not define itself).

Axis	Field Name	Layer	Normative Requirement
6	`author`	Testimony	Identity of the party who posted the entry, as a matter of record.
7	`reason`	Testimony	Stated purpose of the movement, in the author's words, at posting time.
8	`consent`	Testimony	Every party touched by the entry must appear; consent is per-entry and per-movement, never blanket.
9	`definition_version`	Testimony	Pinned registry version of the terms used; entries are read under their own version, and semantic drift is flagged on read.
10	`momentum`	Deal (Ijiri)	Declared trajectory: projected size of the next movement along the chain; breaks are refused except for Axis 3 forgiveness.
11	`receipt`	Deal (Grigg)	Countersigned canonical SHA-256 record held in common by both parties; private alteration fails verification.

3.1 Why Forgiveness Is Axis 3 (and Not an Application Feature)

> *"The third dimension of bookkeeping is forgiveness." — Topher Duguay (October 24, 2026)*

1. **The Spatial Triad of Obligation:** Debit and Credit define a 2D plane of bilateral transfer. Without a third axis specifying the standing of that transfer (`owed | settled | forgiven`), a book can accumulate obligations but cannot close them honestly without faking a payment; disputes then happen off-book.
2. **Append-Only Grace (Reversal Without Erasure):** Forgiveness posts a new append-only reversal entry with `forgiveness = 'forgiven'`, citing the original `owed` entry as its provenance, reversing debit and credit exactly, and carrying both parties' signatures. Nothing is erased; grace is auditable.
3. **Refusing Counterfeit Grace:** A forgiveness entry is valid only against an entry standing `owed`. Attempting to 'forgive' an already `settled` entry in order to rewrite or claw back a completed transfer is deterministically refused.
4. **First in Architecture, Last in Bureaucratic Discovery:** When bureaucracies attempt to cure fraud through procedure alone, they bolt on Axes 5–11 first (provenance, signatures, reasons, consent forms, definitions, trajectory audits, and receipts). Yet *a book with every axis except forgiveness is still Hell—just a well-documented one.* Placing Forgiveness at Axis 3 establishes its structural priority, including its role as the sole lawful exception to Axis 10 momentum.

4. Entry Lifecycle

Entries are append-only (`posted_at` serves as the temporal index surface). Standing transitions move from `owed` to `settled` via a subsequent payment entry, or from `owed` to `forgiven` via an exact reversal entry. When an old entry is read under a newer `definition_version`, term diffs are flagged immediately rather than silently merged.

5. The Ten Deterministic Checker Rules (Implemented in ledger.py)

Rule	Axis	Deterministic Checker Requirement	Fraud / Lie Refused
1	ID	Refuse duplicate entry_id.	Silent history overwrite.
2	Axes 1–2	Refuse unregistered debit or credit accounts.	Ghost/shell account routing.
3	Axis 4	Refuse non-positive amounts (amount <= 0).	Sign-flip direction laundering.

Rule	Axis	Deterministic Checker Requirement	Fraud / Lie Refused
4	Axis 3	Standing must be owed, settled, or forgiven.	Ambiguous obligation status.
5	Axis 3	**Counterfeit Grace Firewall:** Forgiveness valid only against an entry standing owed, reversing (debit, credit, amount) exactly.	Clawing back a settled transfer or altering amounts under the guise of mercy.
6	Axis 5	Provenance chain must walk back and terminate at GENESIS without cycles or missing links.	Invented history (amnesia) or circular self-definition.
7	Axis 8	Consent tuple must include every party touched by debit and credit.	Fiat theft with perfect arithmetic.
8	Axis 9	definition_version must be registered; flag semantic drift across versions on read.	Mid-contract euphemism and definition drift fraud.
9	Axis 10	Child amount must match parent's declared momentum, except when Axis 3 is forgiven.	Trajectory bait-and-switch over time.
10	Axis 11	Common receipt SHA-256 signature must verify for every touched party.	Post-signing alteration of one party's private book.

6. Proof of Concept (11d-spreadsheet-poc)

A zero-dependency Python reference implementation accompanies this paper in the `11d-spreadsheet-poc` directory of `sweetbrier-node` (`ledger.py`, `demo.py`, `README.md`). In the demonstration across three Winnipeg food-bank organizations (Harvest Manitoba, NorWest, and Siloam Mission), four honest entries post and hold—including a 10-crate standing `owed` obligation (`E3`) and its `forgiven` reversal (`E3-forgiven`)—while reading `E1` across drifted definitions (`v1` to `v2`) flags the redefined term `'surplus'`, and seven malformed entries (`B1`–`B6`) are each deterministically refused on a different axis. The POC is an executable specification aid, not a production database; its credibility rests on its exact scope.

7. Relation to ERC-8004

The eleven axes are proposed as an entry-level trust addendum to `ERC-8004`: any validation, reputation, or task-settlement entry in the `ERC-8004` stack can carry the extended eleven-entry schema anchored by its Axis 11 receipt hash. Nothing in this paper constrains what agents may be built or what they may transact. It constrains only what a completed entry may claim about itself.

8. Immediate Applications

- **Mutual-Aid & Supply-Chain Coordination (One Big Backend Food Module):** Coordinating surplus across 18 Winnipeg organizations with counts-only public views, full provenance chains, pinned food-grade definitions, and formal debt forgiveness.
- **Ad Hoc Artist & Patron Syndicates (One Big Consortium):** Pre-work deterministic Split Sheets inside an unincorporated association where every participant receives every receipt and projects requiring lying about splits, receipts, or provenance cannot be sheeted.
- **Autonomous Agent Service Markets:** Protecting agent registries where reputation must survive definition drift and multi-step trajectory manipulation.

9. Limitations and Open Problems

Definition registries (`Axis 9`) require community governance; this specification assumes a versioned registry exists. Momentum (`Axis 10`) is currently checked as a scalar projection; statistical trajectory envelopes are future work. Cryptographic key management and identity binding are inherited from the host `ERC-8004` / `EIP-712` environment.

10. Summary of the Safety Claim

To post a lie into an Eleven-Entry Book, a poster must forge a terminating provenance chain, collect the explicit consent of everyone it names, pin meanings to a registered dictionary version, match a trajectory declared in advance (or obtain bilateral signatures for genuine forgiveness), and hold a receipt identical to their counterparty's. **At that point, the distinction between forging the book and keeping it honestly has quietly disappeared.**